TECHNICAL NOTE

TN104

IRS COMMAND INTERFACE

REV: 2.1

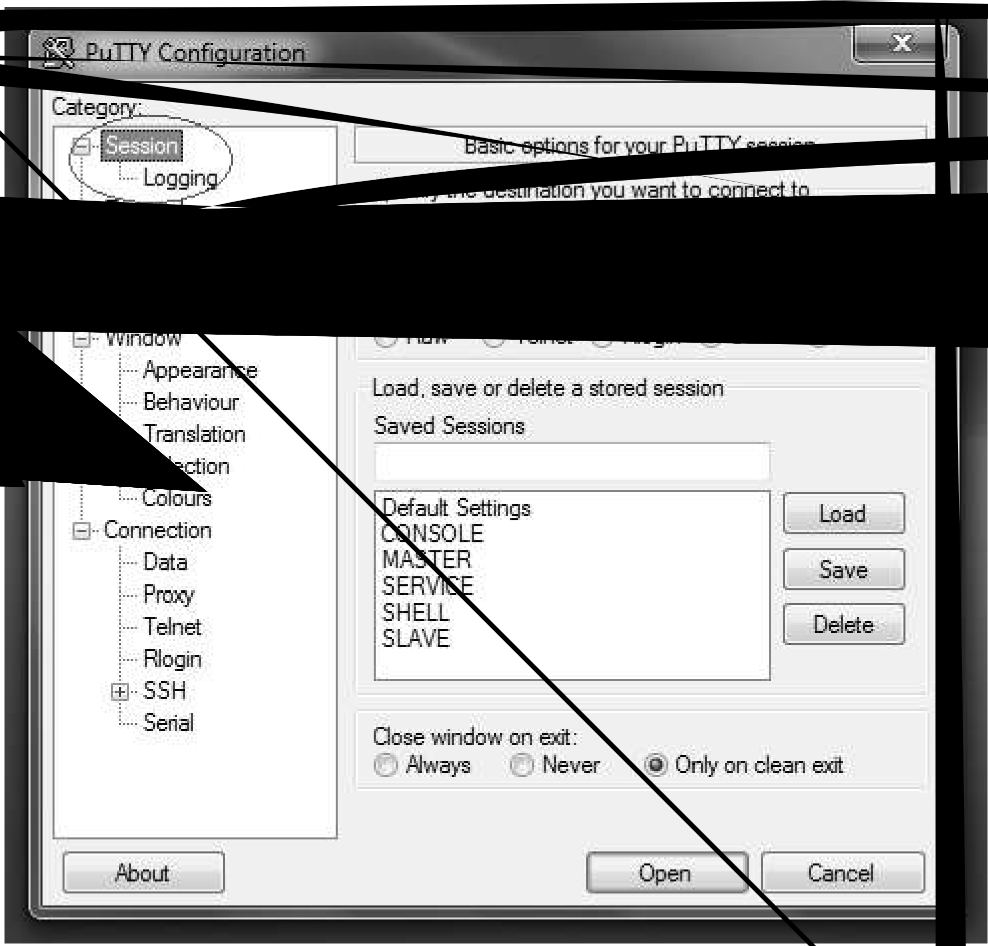
REVISION HISTORY

Rev	Date	Change Description	Prepare		
		pplicable to Z175/ID15	M. Rispoli .	L.Nastasi .	G.B.Peretta .
2.1	07/09/2026	Applicable to Z175/ID16	M. Rispoli .	L.Nastasi .	G.B.Peretta .

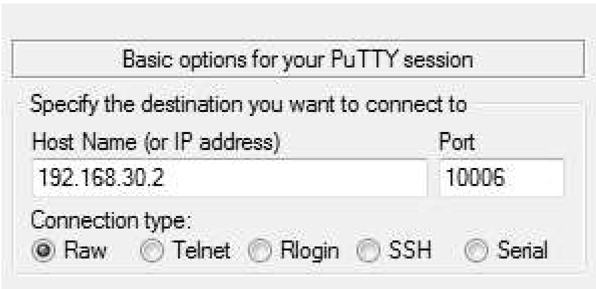
CHANGE LOG

- Compressor: Added command to set the minimum selectable target compression;
- Config: Added command to change the compressor release behavior
- System: Added a command to test the SSR relay integrity;
- Biopsy: Added command to enable/disable the use of the Y upright position sensor;

TABLE OF CONTENT



3. Set the "Basic options.." fields as follows:



4. Press the button OPEN to start the ses~~sion: If the connection success~~ [redacted]e a

| system:
| system >

In the example above, the "system" sub group is entered, having access to a set of commands belonging to this group.

In order to get the command content of the group, it can be send the **?** command into the group:

| system: ?

```
NOTE: when a group is entered it is not necessary to repeat the group name w
```

In order to get the whole group availables, the user can type the command **?** in the root of the Terminal:

Type the following commands:

| ..
| ?

The ".." command causes the Terminal enter the MAIN group (the root of all groups);

The command "?" request for the list of all availab ups

7. Group command description

7.1. SYSTEM COMMANDS (system:)

7.1.1. ***setDATE***

COMMAND
setDATE Year Month

are revisions in the package

COMMAND
Set the Tomo calibration Mode

NOTE: the study is automatically Opened

7.1.8. ***setOperatingMode***

COMMAND
setOperatingMode
DESCRIPTION
Set the Operating mode

NOTE: the study is NOT automatically Opened

7.1.9. ***reboot***

COMMAND
reboot
DESCRIPTION
The Master and Slave Terminals will reboot

7.1.10. ***setPowerOff***

COMMAND
setPowerOff
DESCRIPTION
Init the System power off

NOTE: this command works only with the Closed Study

7.1.11. ***setUnpark***

COMMAND
setUnpark
DESCRIPTION
Activate the ARM unparking procedure

7.1.12. ***setPark***

the Filter selection.

d Filter (no filter selection).

COMMAND
DESCRIPTION
Set the Gantry serial number

NOTE: The serial_number shall be composed of only digits

Usage:

> ***setSN 123456***

set the Gantry serial number to 123456

> ***setSN .***

erases the serial number!

7.2.8. ***setPSW***

COMMAND
setPSW password_number
DESCRIPTION
Set the Gantry service password number

NOTE: The password_number shall be composed of onl

to OFF!!

7.2.10. ***setTubeTemp***

COMMAND
setTubeTemp LT HT
DESCRIPTION
Set the Tube alarm threshold(HT) and the reset alarm threshold (LT)

- HT: is the Tube temperature in (°C) to trigger the Alarm;
- LT: is the Tube temperature in (°C) to reset the Alarm;

7.2.11. ***setLanguage***

COMMAND
setLanguage [ITA/ENG/SPA/FRA/POR/RUS]
DESCRIPTION
Set the GUI current language

NOTE: The Language is immediatelly changed!!

7.2.12. ***resetGantry***

COMMAND
resetGantry
DESCRIPTION
Clear the Gantry System Configuration file

Use this command when the Gantry shall be reconfigured for the hardware setup:

- Motor/Manual rotation;
- HS or LS starter;
- Tilting presence;

7.2.13. ***resetKvCalib***

COMMAND
resetKvCalib
DESCRIPTION
Clear the kV-sense calibration file

Use this command when the kV read back calibration shall be removed.

NOTE: In case of switch between 35kV generator to 49kV or viceversa it is recommended to clear the calibration file!!

7.2.14. ***sysBackup***

COMMAND
sysBackup backup_name
DESCRIPTION
Create a backup of both Master and Slave terminals

The command creates the backup files of the Master terminal and the Slave terminal, stored in the HOME directories:

- MASTER: /home/user/master_***backup_name***.tar
- SLAVE: /home/user/slave_***backup_name***.tar

7.2.

The command doesn't update the Devices. Use the download command (see below)

7.3.1.2. ***storeColliConf***

COMMAND
storeColliConf
DESCRIPTION
Stores the Collimator configuration file

7.3.1.3. ***updateColliU1***

COMMAND
updateColliU1
DESCRIPTION
Updates the microcontroller U1 with the current calibration parameters related to that microcontroller

7.3.1.4. ***updateColliU2***

COMMAND
updateColliU2
DESCRIPTION
Updates the microcontroller U2 with the current calibration parameters related to that microcontroller

7.3.2. 2D Paddle Collimation calibration

7.3.2.1. ***setCalib2D***

COMMAND
setCalib2D Paddle Focus Left Right Front Back Trap
DESCRIPTION
Set the Paddle collimation parameters

- Paddle is one of the following paddles type:
    - PAD24x30
    - PAD18x24_C
    - PAD18x24_L
    - PAD18x24_R
    - PADBIOP_3D
    - PAD9x21

[redacted] ition: 0 to 255;
[redacted] ap collimation blade position: 0 to 255;

[redacted] on Format activation

[redacted] ands allow to control the Collimator Device bypassing the Gantry

[redacted]

[redacted] Collimator Device activation mode

[redacted] cription:

[redacted] : ASSY 01 or ASSY 02;
[redacted] NE LAME: MANUALE/AUTMATICA;
[redacted] NE FILTRO: MANUALE/AUTOMATICA;

When the Collimation type is AUTOMATICA then the Gantry controls the collimation.
When the Collimation type is MANUALE then the collimation is controlled manually by IRS commands.

7.3.3.2. ***setManual***

COMMAND
setManual
DESCRIPTION
Set Only the collimation format in Manual mode

The Filter setting remains in Automatic mode!!

7.3.3.3. ***setAuto***

COMMAND
setAuto
DESCRIPTION

COMMAND
setF val

DESCRIPTION
the front manual blade position

sition is valid only in Manual Mode and it is updated only after the command -

and it is updated only after the command -

[redacted]

command storeColliConf to update [redacted]

7.3.5. Dynamic Filter activation calib[redacted]

This section controls the filter position follower when [redacted]

7.3.5.1. ***getTomoFilterConfig***

COMMAND
getTomoFilterConfig
[redacted]CRIPTION
[redacted] filter parameters

[redacted]

- *Inseguimento filtro: abilitato* -> The dynamic filter is activated;
- *Inseguimento filtro: disabilitato* -> The dynamic filter is not active;
- *Correzione posizione: xxx* -> This is the current Adjust position value;
- *Angoli avanzamento: chg0, chg1, chg2, chg3, chg4, chg5* -> This is the current change angle vector set;

7.3.5.2. ***enableTomoFilter***

COMMAND
enableTomoFilter param
DESCRIPTION

Acivates/Deactivates the filter dynamic feature

- param = ON -> Activates the dynamic filter;
- param = OFF -> Deactivates the dynamic filter;

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

7.3.5.3. ***adjustNominalFilterPosition***

COMMAND
adjustNominalFilterPosition adj
DESCRIPTION

sets the position correction value applied to the initial filter position

- adj: signed value (positive/negative) changing the initial position of the filter.

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

7.3.5.4. ***setFilterChangeAngles***

COMMAND
setFilterChangeAngles chg0 chg1 chg2 chg3 chg4 chg5
DESCRIPTION

sets the array of the filter change angles, in order from negativ to positive

- chg0 to chg5: signed value, in the range of [-27 to 27]

NOTE: the configuration file and the device microcontroller U2 are updated after this command!

7.

List

DESCRIPTION

Returns the list of available compressor paddles

7.4.3.2. *setCalibPad*

COMMAND

setCalibPad offset k-force weight

DESCRIPTION

Changes the current detected paddle calibration parameters

This command allow to tune the current detected paddle parameter in order to provide a more accurate breast thickness measure.

7.4.3.3. *setThi*

no more) a phantom.

...sure the true thickness setting it

...rent paddle elasticity

...n this calibration the User:

- ...ess to 150N a phantom;
- ...e the true thickness at this compression;
- ...the kF_val until the detected thickness equal the measured thickness.

...on Force Parameters

...cted compression force

...n.

COMMAND
Set the minimum selectable Automatic target compression

The value can be set in the range: 30 to 200 Newton.

7.5. MOTOR ACTIVATION COMMANDS (rotazioni:)

7.5.1. Configuration File Management

7.5.1.1. ***readTrxConfig***

COMMAND
readTrxConfig
DESCRIPTION
Read the TRX co

fig
DESCRIPTION
Store the ARM configuration File

7.5.1.5. ***readLenzeConfig***

COMMAND
readLenzeConfig
DESCRIPTION
Read the LENZE configuration File

7.5.1.6. *saveLenzeConfig*

COMMAND
saveLenzeConfig
DESCRIPTION
Store the LENZE configuration File

7.5.2. Inclinome

COMMAND
DESCRIPTION
Activate the TRX to a target angle

The target can be of different nature:

- STOP: stops any current rotation;
- WHOME: move TRX to Wide Home position;
- NHOME: move TRX to Narrow Home position;
- IHOME: move TRX to Intermediate Home position;
- WEND: move TRX to Wide End position;
- NEND: move TRX to Narrow End position;
- IEND: move TRX to Intermediate End position;
- angle: the target is in (°). Example: TRX 10 moves TRX to 10°

7.5.3.2. ***TRX LOOP***

COMMAND
TRX LOOP angle
DESCRIPTION
Activate the TRX in a loop mode

The TRX will move from +angle and -angle until a STOP command is provided.

7.5.3.3. ***ARM***

COMMAND
ARM target
DESCRIPTION
Activate the ARM to a target angle

7.6. POTTER COMMANDS (potter:)

7.6.1. ***setGrid2D***

COMMAND
setGrid2D ON/OFF
DESCRIPTION
Activate (ON) or Deactivate (OFF) the 2D grid (if present)

7.6.2. ***setGrid3D***

COMMAND
setGrid3D ON/OFF
DESCRIPTION
Activate (ON) or Deactivate (OFF) the 3D grid (if present)

7.7. AWS SIMULATOR COMMANDS (aws:)

7.7.1. ***simulateRx***

COMMAND
simulateRx frame
DESCRIPTION
Simulate the reception of an acquisition command frame

The frame shall be a valid command frame, in the format:
<ID NUM %Command param ... %>

7.7.2. ***getFormat***

COMMAND
getFormat frame
DESCRIPTION
Returns information about the frame format

The command can be used to test the validity of a frame
that should be simulated.

7.7.3. ***pcPowerOff***

COMMAND
pcPowerOff frame
DESCRIPTION
Requests to the PC the permission to activate a Power Off sequence

The command can be used to test the validity of a frame
that should be simulated.

7.8. NEW BIOPSY-AL COMMANDS (biopsy:)

The following commands refer to the Biopsy device with the lateral Approach.

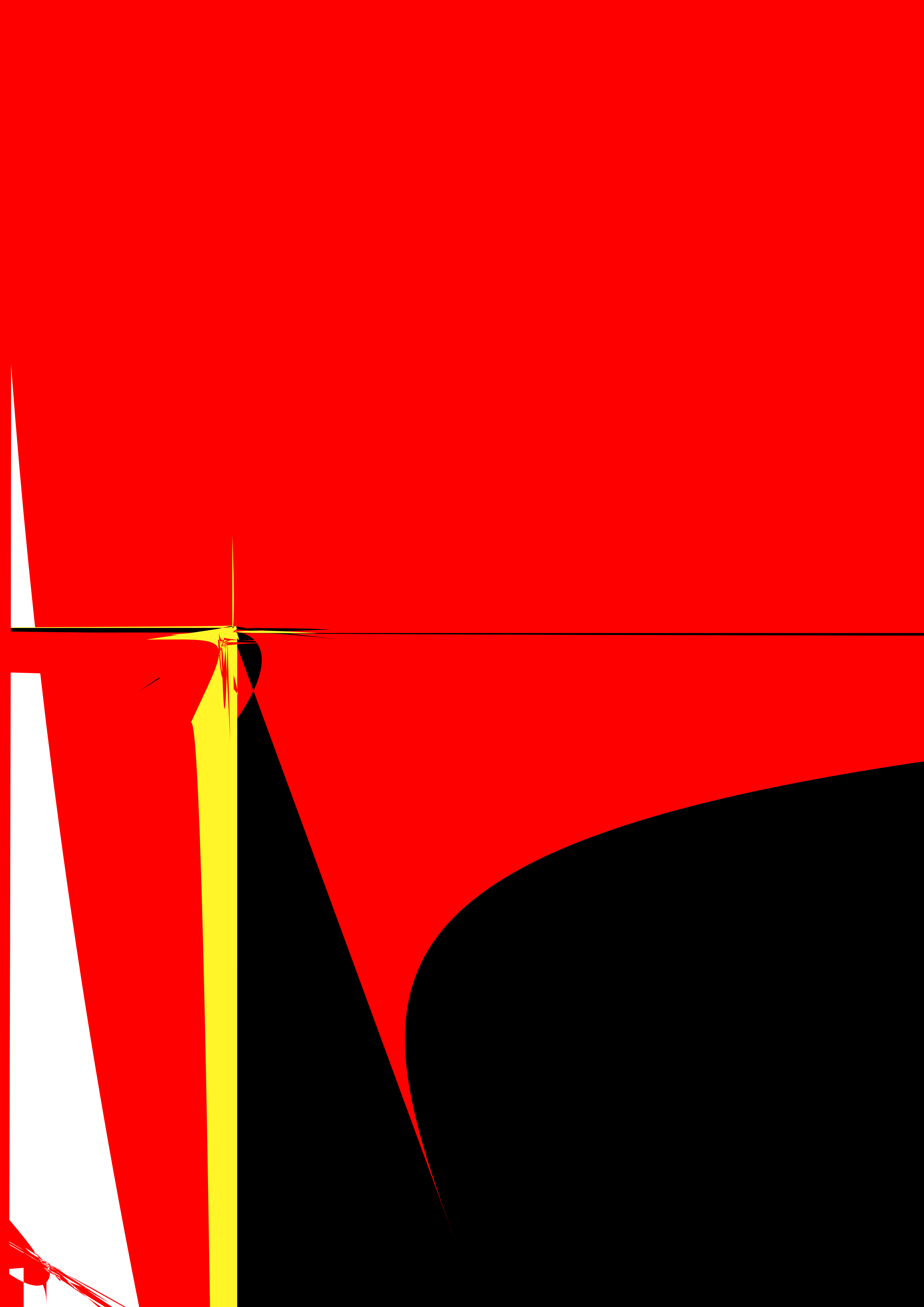
[redacted]nquiry commands

[redacted]etAdapter

[redacted]AND
[redacted]

.1.3. *getSignals*

COMMAND
getSignals
DESCRIPTION
Returns the sens



7.9.3.2. ***activateManualXray***

COMMAND
activateManualXray tubeType
DESCRIPTION
Activate the Manual Exposure session uploading the Tube template type

tubeType parameter:

- tubeType = "35KV": upload the TEMPLATE_35KV_XM1016T configuratin file;
- tubeType = "49KV": upload the TEMPLATE_XM1016THI configuratin file;

When the session is open the Study is Open and the Gantry is ready to expose.

The initial exposure parameters are:

- kV = 20;
- Focus = WL;
- mAs = 10;
- Idac = 2000;

7.9.3.3. ***setExposure***

COMMAND
setExposure Focus kV mAs Idac
DESCRIPTION
Set the next exposure data

Focus parameter:

- Focus = "WL": will use the W large focus;
- Focus = "WS": will use the W small focus;
- Focus = "MoL": will use the Mo large focus;
- Focus = "MoS": will use the Mo small focus;

kV parameter: set 20 to 49 kV (max 35 for 35 KV template file);

mAs parameter: set in a range 1 t

COMMAND

Clos

7.10.2.

COMMAND

COMMAND
Send a Command frame to the ***target***

the target can be:

- PCB269: this is the Compressor board;
- PCB190: this is the Generator board;
- PCB244: this is the Potter board;
- PCB249U1: this is the Collimator U1;
- PCB249U2: this is the Collimator U2;

command_code and command_param can be in hex format or decimal format;

7.11.1.8. ***special***

COMMAND
special *target b1 b2*
DESCRIPTION
Send a Special frame to the ***target***

the target can be:

- PCB26[illegible] this is the Compressor board;
- PCB190: this is the Generator board;
- PCB244: this is the Potter board;
- PCB249U1: this is the Collimator U1;
- PCB249U2: this is the Collimator U2;

b1 and b2 can be in hex format or decimal format;

7.11.2. Debug Terminals

There are two possible debugging terminal types:

- The RS232 serial terminal;
- The TcpIp terminal.

7.11.2.1. The RS232 serial terminal

The Gantry provides a DB9 connector in the back side of the cabinet providing two serial link:

- A serial link for low level debugging in the Master Terminal;
- A serial link for the low level debugging in the Slave Terminal;

After Gantry startup, those terminals remain active only for the system configuration setup time (the early steps of the Gantry startup) then they are disabled.

In order to activate the serial links the user shall
use the following a IRS command:

- debug: ***enablePrint***

After the command is used both serial links will be activated until
the system Power Off.

7.11.2.2. The TcpIp terinal debug

The user can open two separated Debug Terminals one for the Master activities and one for the Slave activities.

The Master terminal connection details are:

- IP: 192.168.30.2;
- PORT: 10010;
- Pro

.168.30.3;

- PORT: 10010;
- Protocol Raw;

In those terminals will be conveyed both the low level activities
as well the high level activities.